

PONY

سلسلة كتب الأستاذ

ICT



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Prepared by:

Gamal Abdelnaser Rezk

Suzy Atef



Lesson 1

- » **External Hard Drive:** Like **HDD**, but much faster, It connects directly to the computer ports.
- » **Ethernet Cable:** It connects a **computer** or **laptop** to a **router**.
- » **Ethernet Port:** It connects an Ethernet cable.
- » **Ethernet** is **more stable** and **faster** than **Wi-Fi**.
- » **Router:** It is a device that connects a computer to the Internet via Ethernet or Wi-Fi.
- » **HDMI:** It transmits **audios** and **videos** from/to devices.

① Bit	The smallest unit of data	0 or 1
② Byte	1 Byte = 8 bits = 1 character	" Cat " = 3 characters = 3 bytes
③ Kilobyte (KB)	1 KB = 1024 B	
④ Megabyte (MB)	1 MB = 1024 KB	
⑤ Gigabyte (GB)	1 GB = 1024 MB	
⑥ Terabyte (TB)	1 TB = 1024 GB	

- » **Megabits per second (Mbps):** It measures the **speed** of a network or the Internet connection.
- » **GigaHertz (GHz):** It measures **the number of cycles** your CPU carries out per second.

Problem	Reason	Solution
① Trouble processing large files (Like a large PowerPoint presentation)	• Not enough graphics storage	• Shut down some programs.

2 Not able to store files	• Not enough space	• Move old files to an external drive.
3 A poor video-conference connection	• Slow wireless Internet connection	• Try an Ethernet connection.
4 The home Internet connection is not working.	• Connection interrupted	• Use the Ethernet or contact your ISP. (Internet Service Provider)

Lesson 2

» **Networks:** They are groups of people or things that are connected to each other for a common purpose.

» **Local Area Network (LAN):**

- It connects computers inside a house or a small office.
- It can also connect printer and router.

» **Internet:** It connects millions of computers from all around the world.

- To connect a computer to the Internet, we need a **gateway**, such as a **router**.

» **Router:** It connects your computer to an ISP to provide Internet access.

» **The ISP:** It is a company that provides the Internet service to the customers.

» **Example of a useful website:** Egyptian Knowledge Bank.

» **Open Network:** It is open and accessible to all, such as (**Internet - WWW**).

» **Closed Network:** It is a private network such as (**Internet – School computer lab**)

Lesson 3

» We use **closed networks** to share **private** information.

» We use **open networks** to share **public** information.

» **Internet of Things (IoT):** can connect digital devices to the Internet using **Wi-Fi**

» IoT allows you to control devices such as Bluetooth speakers from anywhere.

» IoT is useful for people of determination.

Lesson 4

» Scientists use office programs, email and video conference to share their discoveries with others.

» **Five steps to solve computer Problems:**

- 1 Identify the problem.
- 2 Plan solutions.
- 3 Try the solutions.
- 4 Review.
- 5 Ask for help.



Problem	Solution
1 The computer slows down when opening an app.	- Shut it down and restart it. - Check for software updates. - Delete unwanted apps.
2 The browser is running slow.	- Check the Wi-Fi signal. - Make sure apps are up-to-date on your device.
3 No Internet connection	- Use the software to troubleshoot your Wi-Fi connection. - Check that the router is connected. - Restart your router. - Try a new Ethernet cable. - Contact the customer service of your ISP for help.
4 Slow to download	- Restart your computer and router. - Check the Internet speed.

Lesson 5

» There are many ways to find information, such as **QR Codes**, **libraries**, the **trusted websites**.

» Input devices such as:

- 1 Keyboard
- 2 Microphone
- 3 Scanner
- 4 Web Camera



»» **Storage Devices** such as:

- ① **Hard Disk Drive (HDD):** The main place for saving files on computer, it has a very large storage capacity.
- ② **Flash Memory:** A small portable device, used to save and transfer files to another device.



»» **File Management:** It means creating folders and putting each type of file in the right place. This makes it easy to find files quickly.

»» **Create the main folder:**

- Press the right click >>> New >>> Folder
- Then, type a name for the folder.

»» A main folder can contain one or more subfolders.



1 Choose the correct answer:

- 1 The smallest unit of data is called
a. Byte b. MB c. Bit d. TB
- 2 Ethernet is than Wi-Fi.
a. slower b. more stable c. faster d. both b and c
- 3 The cable that connects a computer to a router is called
a. HDMI b. Wi-Fi c. Ethernet cable d. ISP
- 4 A device that connects computers to the Internet through Wi-Fi or Ethernet is called
a. Modem b. Router c. Flash memory d. CPU
- 5 1 Kilobyte = Bytes
a. 1024 b. 100 c. 512 d. 64
- 6 The unit used to measure Internet speed is
a. GHz b. MB c. Mbps d. GB
- 7 The device that stores and transfers files between computers is
a. scanner b. flash memory
c. router d. monitor
- 8 The correct order of data units from smallest to largest is
a. Bit – Byte – KB – MB – GB – TB
b. Byte – Bit – MB – GB – TB
c. KB – Bit – MB – TB – GB
d. TB – GB – MB – KB – Byte
- 9 A group of connected computers and devices for a common purpose is called
a. Internet b. database c. network d. router
- 10 A LAN connects computers within
a. a home b. small office c. the world d. both a and b

- 11 is the global network that connects millions of computers worldwide.
 a. LAN b. intranet c. Internet d. gateway
- 12 The company that provides Internet services to users is called
 a. ISP b. CPU c. HDD d. LAN
- 13 An open network that anyone can access is
 a. intranet b. closed network
 c. Internet d. Private LAN
- 14 A school computer lab network is an example of
 a. open network b. closed network
 c. Internet d. global system
- 15 Internet of Things (IoT) allows you to connect digital devices using
 a. Wi-Fi b. USB c. Ethernet d. Bluetooth only
- 16 Businesses use networks to share private information.
 a. open b. closed c. public d. global
- 17 Intranet is network.
 a. open b. closed c. private d. both b and c
- 18 The first step in solving computer problems is to
 a. ask for help b. identify the problem
 c. review d. try solutions
- 19 A solution for a browser running slowly is to
 a. check Wi-Fi signal b. replace monitor
 c. buy printer d. delete folder
- 20 If there is no Internet connection, you can
 a. restart router b. contact ISP
 c. try new Ethernet cable d. all of them
- 21 The unit "GHz" is used to measures
 a. file size b. CPU speed
 c. network bandwidth d. Internet data
- 22 A drive like HDD but faster and external is called
 a. flash b. external hard drive
 c. modem d. SSD cable

- 23 A video conference may be poor because of
 a. slow Wi-Fi b. no RAM c. keyboard error d. router speed
- 24 To solve slow download problems, you can
 a. restart the router b. check Internet speed
 c. restart computer d. all of them
- 25 When unable to store files, the best solution is to
 a. delete apps b. move old files to external drive
 c. restart d. turn off Wi-Fi
- 26 is a cable used to transmit audio and video.
 a. Ethernet b. Wi-Fi c. HDMI d. both a and b
- 27 is very fast and like a USB or SSD.
 a. HDMI b. network c. port d. flash memory
- 28 The is used to insert images and use them later.
 a. keyboard b. printer c. microphone d. scanner
- 29 You can organize files on your device using
 a. Folders b. Copy c. Paste d. Limited space
- 30 We can create a main folder and add to it.
 a. sub-folders b. printer c. delete folder d. speakers

2 Put (✓) or (X):

- | | |
|---|-------|
| 1 Bit is the smallest unit of data. | () |
| 2 Ethernet is faster and more stable than Wi-Fi. | () |
| 3 HDMI cable connects computers to the router. | () |
| 4 One kilobyte equals 1024 bytes. | () |
| 5 The word "ICT" takes 3 bytes to store. | () |
| 6 A router connects multiple devices to the Internet. | () |
| 7 Mbps measures CPU performance. | () |
| 8 LAN connects computers within a home or office. | () |
| 9 The Internet is an example of a closed network. | () |
| 10 The school computer lab is a type of intranet. | () |

- 11 The ISP provides Internet service to customers. ()
- 12 IoT allows devices to connect through Bluetooth and Wi-Fi. ()
- 13 Internet of Things is not useful for people of determination. ()
- 14 Scientists use Word, Excel, and PowerPoint in their research. ()
- 15 Restarting the router can fix "no Internet" problems. ()
- 16 Checking Wi-Fi signal helps when the browser runs slow. ()
- 17 The first step in problem solving is to review the solution. ()
- 18 You can contact your ISP when Internet is down. ()
- 19 Restarting the computer can solve slow app issues. ()
- 20 You should ignore software updates to keep computer stable. ()
- 21 A flash memory cannot transfer data to another device. ()
- 22 Ethernet cable is connected in devices to holes called ethernet port. ()
- 23 Closed networks are accessible to everyone. ()
- 24 Internet of Things connects devices using wires only. ()
- 25 Using Ethernet make video conferencing more stable. ()
- 26 A QR code is used to get information quickly. ()
- 27 The main folder can contain one or more subfolders. ()
- 28 It is preferable to give each folder an appropriate name. ()
- 29 To create a new folder, left-click on an empty space. ()
- 30 An operating system does not allow you to add a sub-folder inside a main folder. ()



Answers

1

1 c

2 d

3 c

4 b

5 a

6 c

7 b

8 a

9 c

10 d

11 c

12 a

13 c

14 b

15 a

16 b

17 d

18 b

19 a

20 d

21 b

22 b

23 a

24 d

25 b

26 c

27 d

28 d

29 a

30 a

2

1 ✓

2 ✓

3 ✗

4 ✓

5 ✓

6 ✓

7 ✗

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14 ✓

15 ✓

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28 ✓

29 ✗

30 ✗